



International Islamic University Chittagong (IIUC)

Department of Computer Science and Engineering (CSE)

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Group A

1. Partial Differential

Q-1. Define partial derivative with example.

Q-2. If $u = a \log(x^2 + y^2) + b \tan^{-1}\left(\frac{y}{x}\right)$ then show that, $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

Q-3. If $v = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$ then show that, $v_{xx} + v_{yy} + v_{zz} = 0$

Q-4. If $u = \log(x^3 + y^3 + z^3 - 3xyz)$ then show that, $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}$

Q-5. Define homogeneous function and State Euler's theorem for homogeneous function.

Q-6. If $u = \sin^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$ then using Euler's theorem show that, $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{\tan u}{2}$

Q-7. If $u = \tan^{-1}\left(\frac{x^3+y^3}{x+y}\right)$ then using Euler's theorem show that, $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$

2. Maxima and Minima

Q-1. Define maxima and minima of a function.

Q-2. When a function will be maxima, minima and neither maxima nor minima?

Q-3. Find the maxima and minima of the function, $f(x) = 2x^3 - 21x^2 + 36x - 20$

Q-4. Find the maxima and minima of the function, $y = 2x^3 - 9x^2 + 12x - 3$

Q-5. Show that the function, $f(x) = x^3 - 3x^2 + 6x + 5$ has neither maximum nor minimum.

3. Indefinite Integral

Method # 02

Q-1. Evaluate the integral $\int \frac{dx}{x^2+4x+13}$

Q-2. Evaluate the integral $\int \frac{dx}{4x^2+8x+13}$

Method # 04

Q-3. Evaluate the integral $\int \frac{x \, dx}{4x^2 + 8x + 13}$

Q-4. Evaluate the integral $\int \frac{3x \, dx}{x^2 - x - 2}$

Q-5. Evaluate the integral $\int \frac{2x+3}{3x^2-x+1} \, dx$

Method # 05

Q-6. Evaluate the integral $\int \frac{x^2-x+1}{x^2+x+1} \, dx$

Q-7. Evaluate the integral $\int \frac{x^2}{x^2-4} \, dx$

Method # 07

Q-8. Evaluate the integral $\int \frac{dx}{(2x+1)\sqrt{4x+3}}$

Q-9. Evaluate the integral $\int \frac{dx}{(2+x)\sqrt{1+x}}$

Method # 13

Q-10. Evaluate the integral $\int x^2 e^x \, dx$

Q-11. Evaluate the integral $\int x \ln x \, dx$

Q-12. Evaluate the integral $\int \tan^{-1} x \, dx$

Method # 14

Q-13. Prove that, $\int e^x \{f(x) + f'(x)\} \, dx = e^x f(x) + c$

Q-14. Evaluate the integral $\int e^x (\sin x - \cos x) \, dx$

Q-15. Evaluate the integral $\int \frac{x e^x}{(x+1)^2} \, dx$

1. Partial Differentiation

▣ A partial derivative is defined as a derivative in which some variables are kept constant and the derivative of a function with respect to the other variable can be determined.

* Example: $f(x, y) = 3x + 4y$

→ Partial differentiating with respect to x
we get,

$$\begin{aligned}\frac{\partial f}{\partial x} &= 3x + 4y \\ &= 3 + 0 \quad [\because y \text{ constant}] \\ &= 3\end{aligned}$$

→ Partial differentiating with respect to y we get,

$$\begin{aligned}\frac{\partial f}{\partial y} &= 3x + 4y \\ &= 0 + 4 \quad [\because \text{here } x \text{ constant}] \\ &= 4\end{aligned}$$

▣ Q-2: Solution

given that,

$$U = a \log(x^2 + y^2) + b \tan^{-1}\left(\frac{y}{x}\right) \text{ --- (i)}$$

Now, Partial differentiating (i) two times with respect to x we get,

$$\frac{\partial U}{\partial x} = a \cdot \frac{1}{x^2 + y^2} \cdot 2x + b \cdot \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \left(-\frac{1}{x^2}\right) \cdot y$$

$$= \frac{2ax}{x^2 + y^2} + \frac{-by}{\frac{x^2 + y^2}{x^2}} \cdot \frac{1}{x^2}$$

$$= \frac{2ax}{x^2 + y^2} + (-by) \times \frac{\cancel{x^2}}{x^2 + y^2} \cdot \frac{1}{\cancel{x^2}}$$

$$= \frac{2ax}{x^2 + y^2} - \frac{by}{x^2 + y^2}$$

$$= \frac{2ax - by}{x^2 + y^2} \text{ --- (ii)}$$

Again,

$$\frac{\partial u}{\partial x} = \frac{2ax - by}{x+y}$$

$$= \frac{(x+y) \cdot 2a - (2ax - by) \cdot 1}{(x+y)^2} \quad \left[\because \frac{d}{dx} \frac{u}{v} \right]$$

$$= \frac{2ax + 2ay - 2ax + by}{(x+y)^2}$$

$$= \frac{2ay + by}{(x+y)^2} \quad \text{--- (iii)}$$

Similarly, Partial differentiating (i) two times with respect to y we get,

$$\frac{\partial^2 u}{\partial y^2} = a \cdot \frac{1}{x+y} \cdot 2 + b \cdot \frac{1}{1 + \frac{y}{x}} \cdot \frac{1}{x}$$

$$= \frac{2ay}{x+y} + b \times \frac{x}{x+y} \cdot \frac{1}{x}$$

$$= \frac{2ay}{x+y} + \frac{bx}{x+y}$$

$$\frac{\partial U}{\partial y} = \frac{2ay + bx}{x+y} \quad \text{--- (iv)}$$

Also,

$$\frac{\partial^2 U}{\partial x \partial y} = \frac{(x+y) \cdot 2a - (2ay + bx) \cdot 2y}{(x+y)^2} \quad \left[\because \frac{\partial}{\partial x} \frac{\partial U}{\partial y} \right]$$

$$= \frac{2ax + 2ay - 4ay - 2bxy}{(x+y)^2}$$

$$= \frac{2ax - 2ay - 2bxy}{(x+y)^2} \quad \text{--- (v)}$$

Now,

L.H.S,

$$\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2}$$

$$= \frac{2ay - 2ax + 2bxy}{(x+y)^2} + \frac{2ax - 2ay - 2bxy}{(x+y)^2}$$

$$= \frac{2ay - 2ax + 2bxy + 2ax - 2ay - 2bxy}{(x+y)^2}$$

$$(ii) \quad z = \frac{0}{(x^v + y^v)^v}$$

$$= 0$$

Hence,

$$(iii) \quad \frac{\partial^2 u}{\partial x^v} + \frac{\partial^2 u}{\partial y^v} = 0$$

[shown]

Q-3: Solution

Given that,

$$(vi) \quad v = \frac{1}{\sqrt{x^v + y^v + z^v}}$$

$$= (x^v + y^v + z^v)^{-\frac{1}{2}} \quad \text{--- (i)}$$

Now, Partial differentiating (i) two times with respect to x we get,

$$\frac{\partial v}{\partial x} = v_x = -\frac{1}{2} (x^v + y^v + z^v)^{-\frac{1}{2}-1} \cdot \cancel{x}$$

$$= -x (x^v + y^v + z^v)^{-\frac{3}{2}}$$

$$\frac{\partial V}{\partial x} = V_x = -x(x^y + y^y + z^y)^{-\frac{3}{2}} \quad \text{--- (ii)}$$

Also,

$$\frac{\partial^2 V}{\partial x^2} = V_{xx} = (-x) \cdot (-\frac{3}{2})(x^y + y^y + z^y)^{-\frac{3}{2}-1} \cdot \cancel{\partial x} - 1 \cdot (x^y + y^y + z^y)^{-\frac{3}{2}} \left[\because \frac{\partial}{\partial x} u^v \right]$$

$$\therefore V_{xx} = 3x^y(x^y + y^y + z^y)^{-\frac{5}{2}} - (x^y + y^y + z^y)^{-\frac{3}{2}} \quad \text{--- (iii)}$$

Similarly, Partial differentiation (i) two times w.r. to y we get,

$$V_y = \frac{\partial V}{\partial y} = -y(x^y + y^y + z^y)^{-\frac{3}{2}} \quad \longrightarrow \text{(iv)}$$

Also,

$$V_{yy} = \frac{\partial^2 V}{\partial y^2} = 3y^y(x^y + y^y + z^y)^{-\frac{5}{2}} - (x^y + y^y + z^y)^{-\frac{3}{2}} \quad \text{(v)}$$

Furthermore,

$$\frac{\partial U}{\partial z} = V_z = -z(x^{\vee} + y^{\vee} + z^{\vee})^{-3/2}$$

Also,

$$V_{zz} = \frac{\partial V}{\partial z^{\vee}} = 3z^{\vee}(x^{\vee} + y^{\vee} + z^{\vee})^{-5/2} - (x^{\vee} + y^{\vee} + z^{\vee})^{-3/2} \quad \text{--- (vi)}$$

L.H.S,

$$\begin{aligned} & V_{xx} + V_{yy} + V_{zz} \\ &= 3(x^{\vee} + y^{\vee} + z^{\vee})(x^{\vee} + y^{\vee} + z^{\vee})^{-5/2} - 3(x^{\vee} + y^{\vee} + z^{\vee})^{-3/2} \\ &= 3(x^{\vee} + y^{\vee} + z^{\vee})^{1 - 5/2} - 3(x^{\vee} + y^{\vee} + z^{\vee})^{-3/2} \\ &= 3(x^{\vee} + y^{\vee} + z^{\vee})^{-3/2} - 3(x^{\vee} + y^{\vee} + z^{\vee})^{-3/2} \\ &= 0 \end{aligned}$$

[shown]

Q-4: Solution

Given that,

$$u = \log(x^3 + y^3 + z^3 - 3xyz) \text{ --- (i)}$$

Now, Partially differentiating (i) with respect to x, y, z respectively,

we get,

$$\frac{\partial u}{\partial x} = \frac{1}{x^3 + y^3 + z^3 - 3xyz} (3x^2 - 3yz)$$

$$= \frac{(3x^2 - 3yz)}{x^3 + y^3 + z^3 - 3xyz}$$

Similarly,

$$\frac{\partial u}{\partial y} = \frac{3y^2 - 3xz}{x^3 + y^3 + z^3 - 3xyz}$$

$$\frac{\partial u}{\partial z} = \frac{3z^2 - 3xy}{x^3 + y^3 + z^3 - 3xyz}$$

Now adding,

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3(x^2 + y^2 + z^2 - xy - yz - zx)}{x^3 + y^3 + z^3 - 3xyz}$$

$$= \frac{3(x^2 + y^2 + z^2 - xy - yz - zx)}{(x+y+z)(x^2 + y^2 + z^2 - xy - yz - zx)}$$

$$= \frac{3}{(x+y+z)}$$

Now,

$$L.S = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) u$$

$$= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) \times \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) \times u$$

$$= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) \times \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} \right)$$

$$= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) \times \frac{3}{(x+y+z)}$$

$$= \frac{\partial}{\partial x} \left(\frac{3}{(x+y+z)} \right) + \frac{\partial}{\partial y} \left(\frac{3}{(x+y+z)} \right) + \frac{\partial}{\partial z} \left(\frac{3}{(x+y+z)} \right)$$

$$= 3 \cdot \frac{-1}{(x+y+z)^2} \cdot 1 + 3 \cdot \frac{-1}{(x+y+z)^2} \cdot 1 + 3 \cdot \frac{-1}{(x+y+z)^2} \cdot 1$$

$$= -9 \cdot \frac{1}{(x+y+z)^2} = R.S \quad [\text{shown}]$$

Q-5: Homogeneous function:-

A homogeneous function is a type of mathematical function that has the same derivative at all points in its domain.

Euler's theorem:-

Euler's theorem is used to establish a relationship between the partial derivatives of a function and the product of the function with its degree.

Formula:
$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = nu$$

Q-6:

Given that,

$$u = \sin^{-1} \frac{x+y}{\sqrt{x}+\sqrt{y}}$$

$$\Rightarrow \sin u = \frac{x+y}{\sqrt{x}+\sqrt{y}}$$

$$= \frac{\cancel{x} (1 + \frac{y}{x})}{\cancel{\sqrt{x}} (1 + \frac{\sqrt{y}}{\sqrt{x}})}$$

$$= \frac{\sqrt{x} (1 + \frac{y}{x})}{(1 + \frac{\sqrt{y}}{\sqrt{x}})}$$

(i) $\therefore \sin u = x^{\frac{1}{2}} \cdot f\left(\frac{y}{x}\right)$ [where, $f\left(\frac{y}{x}\right) = \frac{1 + \frac{y}{x}}{1 + \frac{\sqrt{y}}{\sqrt{x}}}$]

Now, using Euler's theorem,

$$x \cdot \frac{\partial}{\partial x} (\sin u) + y \cdot \frac{\partial}{\partial y} (\sin u) = \frac{1}{2} \cdot \sin u$$

$$\Rightarrow x \cos u \frac{du}{dx} + y \cos u \frac{du}{dy} = \frac{1}{2} \sin u$$

$$\Rightarrow \cos u \left(x \frac{du}{dx} + y \frac{du}{dy} \right) = \frac{1}{2} \sin u$$

$$\Rightarrow x \frac{du}{dx} + y \frac{du}{dy} = \frac{1}{2} \cdot \frac{\sin u}{\cos u}$$

$$\therefore x \frac{du}{dx} + y \frac{du}{dy} = \frac{1}{2} \tan u \quad [\text{showed}]$$

Q-7:

Given that,

$$U = \tan^{-1} \left(\frac{x^3 + y^3}{x + y} \right)$$

$$\Rightarrow \tan U = \frac{x^3 + y^3}{x + y}$$

$$\Rightarrow \tan U = \frac{x^3 (1 + \frac{y^3}{x^3})}{x (1 + \frac{y}{x})}$$

$$= x^2 \left\{ \frac{1 + (\frac{y}{x})^3}{1 + \frac{y}{x}} \right\}$$

$$\therefore \tan U = x^2 f\left(\frac{y}{x}\right) \text{--- (i)}$$

By using Euler's theorem,

$$x \cdot \frac{\partial}{\partial x} (\tan U) + y \frac{\partial}{\partial y} (\tan U) = 2 \tan U$$

$$\Rightarrow x \cdot \sec^2 U \frac{\partial U}{\partial x} + y \sec^2 U \frac{\partial U}{\partial y} = 2 \frac{\sin U}{\cos U}$$

$$\Rightarrow \sec^2 U \left(x \frac{\partial U}{\partial x} + y \frac{\partial U}{\partial y} \right) = 2 \frac{\sin U}{\cos U}$$

$$\Rightarrow x \frac{\partial U}{\partial x} + y \frac{\partial U}{\partial y} = 2 \cdot \frac{\sin U}{\cos U} \cdot \frac{1}{\sec^2 U}$$

$$\Rightarrow x \frac{\partial U}{\partial x} + y \frac{\partial U}{\partial y} = 2 \cdot \frac{\sin U}{\cos U} \cdot \cos^2 U$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2 \cdot \frac{\sin u}{\cos u} \cdot \cancel{\cos u} \cos u$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2 \sin u \cdot \cos u$$

$$\therefore x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$$

Q-8:

Maxima and Minima

$\downarrow$ $\downarrow$
 Maximum & Minimum

* Maximum & Minimum: Maxima and minima are known as the extrema of a function.

Maxima and minima are the maximum or the minimum value of a function within the given sets of ranges.

#Q-2: Condition:

- i) The function $y = f(x)$ will be maximum at $x = a$, when $f'(a) = 0$ and $f''(a) < 0$.
- ii) The function $y = f(x)$ will be minimum at $x = a$ when $f'(a) = 0$ and $f''(a) > 0$.
- iii) The function $y = f(x)$ has neither maximum nor minimum at $x = a$ if $f'(a) \neq 0$.

#Q-3:

Given that,

$$f(x) = 2x^3 - 21x^2 + 36x - 20$$

1st differentiation $\therefore f'(x) = 6x^2 - 42x + 36$

2nd differentiation $\therefore f''(x) = 12x - 42$

Now, for Maxima and minimum values,

$$f'(x) = 0$$

$$\Rightarrow 6x^2 - 42x + 36 = 0$$

$$\Rightarrow x^2 - 7x + 6 = 0$$

$$\Rightarrow x^2 - 6x - x + 6 = 0$$

$$\Rightarrow x(x-6) - 1(x-6) = 0$$

$$\Rightarrow (x-1)(x-6) = 0$$

$$\therefore x = 1, 6$$

Now, when, $x = 1$ then $f'(1) = 12 \cdot 1 - 42$

$$= -30 < 0$$

So, at $x = 1$ the given function will be Maxima and the maximum value is

$$f(x) = 2x^3 - 21x^2 + 36x - 20$$

$$f(1) = 2 \cdot 1^3 - 21 \cdot 1^2 + 36 \cdot 1 - 20$$

$$= 2 - 21 + 36 - 20$$

$$= 38 - 41$$

$$= -3$$

Again,

$$\text{when } x=6 \text{ then } f''(6) = 12 \cdot 6 - 42 \\ = 30 > 0$$

So, at $x=6$ then given function will be minima and the minima value is

$$f(x) = 2x^3 - 21x^2 + 36x - 20$$

$$\therefore f(6) = 2 \cdot 6^3 - 21 \cdot 6^2 + 36 \cdot 6 - 20$$

$$= -128$$

Q-4:

Given that,

$$y = 2x^3 - 9x^2 + 12x - 3$$

$$y = f(x) = 2x^3 - 9x^2 + 12x - 3$$

$$\text{1st differentiation } \Rightarrow f'(x) = 6x^2 - 18x + 12$$

$$= 6(x^2 - 3x + 2)$$

$$\text{2nd differentiation } \Rightarrow f''(x) = 12x - 18$$

Now, for Maxima and minima values,

$$f'(x) = 0$$

$$\Rightarrow 6(x^2 - 3x + 2) = 0$$

$$\Rightarrow x^2 - 2x - x + 2 = 0$$

$$\Rightarrow x(x-2) - 1(x-2) = 0$$

$$\Rightarrow (x-2)(x-1) = 0$$

$$\therefore x = 1, 2$$

$$\begin{aligned} \text{Now, when } x=1 \text{ then } f''(\cdot) &= 12 \cdot 1 - 18 \\ &= -6 < 0 \end{aligned}$$

So, at $x=1$ the given function will be Maxima and the maxima value is

$$y = f(x) = 2x^3 - 9x^2 + 12x - 3$$

$$\therefore f(1) = 2 \cdot 1^3 - 9 \cdot 1^2 + 12 \cdot 1 - 3$$

$$= 2 - 9 + 12 - 3$$

$$= 2$$

Again,

$$\text{when } x=2 \text{ the } f''(2) = 12 \cdot 2 - 18 \\ = 6 > 0$$

So, at $x=2$ the given function will be minima and the minima value is

$$y = f(x) = 2x^3 - 9x^2 + 12x - 3$$

$$\therefore f(2) = 2 \cdot 2^3 - 9 \cdot 2^2 + 12 \cdot 2 - 3 \\ = 16 - 36 + 24 - 3 \\ = 1$$

Answer.

#Q-5: Given that,

$$f(x) = x^3 - 3x^2 + 6x + 5$$

$$\therefore f'(x) = 3x^2 - 6x + 6$$

$$= 3(x^2 - 2x + 2)$$

$$= 3(x^2 - 2x + 1 + 1)$$

$$= 3\{(x-1)^2 + 1\}$$

which is always non-zero for any real values of x
hence, the given function has neither Maxima
nor minima.

Indefinite Integral

NOTE: $\int f(x) dx = g(x) + C$ $\rightarrow$ const of Integration.

$\swarrow$ Integral $\downarrow$ Integrand

Method # 02:-

Q-1: let,

$$I = \int \frac{dx}{x^2 + 4x + 13}$$

$$= \int \frac{dx}{x^2 + 2 \cdot x \cdot 2 + 2^2 + 13 - 2^2} \quad [\because 2^2 \text{ Extra add/যোগ করলাম}]$$

$$= \int \frac{dx}{(x+2)^2 + 9}$$

$$= \int \frac{dx}{(x+2)^2 + 3^2}$$

$$= \frac{1}{3} \tan^{-1} \frac{x+2}{3} + C \quad \left[\because \int \frac{dx}{a^2 + x^2} = \right.$$

$$\left. \tan^{-1} \frac{x}{a} + C \right]$$

8-2:

let,

$$I = \int \frac{dx}{4x^2 + 8x + 13}$$

$$= \int \frac{dx}{4(x^2 + 2x + 13/4)}$$

$$= \frac{1}{4} \int \frac{dx}{x^2 + 2 \cdot x \cdot 1 + 1^2 + 13/4 - 1}$$

$$= \frac{1}{4} \int \frac{dx}{(x+1)^2 + (3/2)^2}$$

$$= \frac{1}{4} \cdot \frac{1}{3/2} \tan^{-1} \frac{x+1}{3/2} + C$$

$$= \frac{1}{6} \tan^{-1} \frac{2(x+1)}{3} + C \quad \text{Ans}$$

$$\left[\because \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C \right]$$

Method # 04 :- $\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$

Q-3:

$$x = \frac{x}{x+1} - 1$$

$$= \frac{1}{8} (8x+8) - 1$$

let,

$$I = \int \frac{x}{4x^2 + 8x + 13} dx$$

$$= \int \frac{\frac{1}{8} (8x+8) - 1}{4x^2 + 8x + 13} dx \quad \left[\because x = \frac{1}{8} (8x+8) - 1 \right]$$

$$= \frac{1}{8} \int \frac{(8x+8) - 8}{4x^2 + 8x + 13} dx$$

$$= \frac{1}{8} \int \frac{(8x+8) dx}{4x^2 + 8x + 13} - \int \frac{1}{4x^2 + 8x + 13} dx$$

$$= \frac{1}{8} \int \frac{(8x+8) dx}{4x^2 + 8x + 13} - \int \frac{1}{4(x^2 + 2x + \frac{13}{4})} dx$$

$$= \frac{1}{8} \int \frac{(8x+8) dx}{4x^2 + 8x + 13} - \frac{1}{4} \int \frac{1}{x^2 + 2x + 1 + \frac{13}{4} - 1} dx$$

$$= \frac{1}{8} \int \frac{(8x+8) dx}{4x^2 + 8x + 13} - \frac{1}{4} \int \frac{1}{(x+1)^2 + \frac{9}{4}} dx$$

$$\begin{aligned}
&= \frac{1}{8} \int \frac{8x+8}{4x^2+8x+13} dx - \frac{1}{4} \int \frac{1}{(x+1)^{\sqrt{}} + (\frac{3}{2})^{\sqrt{}}} dx \\
&= \frac{1}{8} \ln |4x^2+8x+13| - \frac{1}{4} \cdot \frac{1}{\frac{3}{2}} \tan^{-1} \frac{x+1}{\frac{3}{2}} + C \\
&= \frac{1}{8} \ln |4x^2+8x+13| - \frac{1}{6} \tan^{-1} \frac{2(x+1)}{3} + C
\end{aligned}$$

Answer

Q-4:

$$\begin{aligned}
&3x \\
&= \left(3x - \frac{3}{2}\right) + \frac{3}{2}
\end{aligned}$$

$$\therefore 3x = \frac{3}{2}(2x-1) + \frac{3}{2}$$

let,

$$I = \int \frac{3x}{x^2-x-2} dx$$

$$= \int \frac{\frac{3}{2}(2x-1) + \frac{3}{2}}{x^2-x-2} dx$$

$$= \frac{3}{2} \int \frac{(2x-1)}{x^2-x-2} + \int \frac{\frac{3}{2}}{x^2-x-2} dx$$

$$= \frac{3}{2} \int \frac{(2x-1)}{x^2-x-2} + \frac{3}{2} \int \frac{1}{x^2-x-2} dx$$

$$= \frac{3}{2} \int \frac{2x-1}{x^2-x-2} + \frac{3}{2} \int \frac{1}{x^2-2 \cdot x \cdot \frac{1}{2} + (\frac{1}{2})^2 - \frac{1}{4} - 2} dx$$

$$= \frac{3}{2} \int \frac{2x-1}{x^2-x-2} + \frac{3}{2} \int \frac{1}{(x-\frac{1}{2})^2 - \frac{9}{4}} dx$$

$$= \frac{3}{2} \int \frac{2x-1}{x^2-x-2} + \frac{3}{2} \int \frac{1}{(x-\frac{1}{2})^2 - (\frac{3}{2})^2} dx$$

$$= \frac{3}{2} \int \frac{2x-1}{x^2-x-2} + \frac{3}{2} \int \frac{dx}{(x-\frac{1}{2})^2 - (\frac{3}{2})^2}$$

$$= \frac{3}{2} \ln|x^2-x-2| + \frac{3}{2} \cdot \frac{1}{\cancel{2} \cdot \frac{3}{2}} \ln \frac{x-\frac{1}{2}-\frac{3}{2}}{x-\frac{1}{2}+\frac{3}{2}} + C$$

$$= \frac{3}{2} \ln|x^2-x-2| + \frac{3}{2} \cdot \frac{1}{3} \ln \frac{x-2}{x+1} + C$$

$$= \frac{3}{2} \ln|x^2-x-2| + \frac{1}{2} \ln \frac{x-2}{x+1} + C$$

Am

$$\int \frac{1}{x^2-a^2} dx = \frac{1}{2a} \ln \frac{x-a}{x+a} + C$$

#Q-5:

let,

$$I = \int \frac{2x+3}{3x^2-x+1} dx$$

$$= \int \frac{\frac{1}{3}(6x-1)+3+\frac{1}{3}}{3x^2-x+1} dx$$

$$= \frac{1}{3} \int \frac{(6x-1) + \frac{10}{3}}{3x^2-x+1} dx$$

$$= \frac{1}{3} \int \frac{6x-1}{3x^2-x+1} dx + \int \frac{10/3}{3x^2-x+1} dx$$

$$= \frac{1}{3} \int \frac{6x-1}{3x^2-x+1} dx + \frac{10}{3} \int \frac{1}{3x^2-x+1} dx$$

$$= \frac{1}{3} \int \frac{6x-1}{3x^2-x+1} dx + \frac{10}{3} \int \frac{1}{3(x^2 - \frac{x}{3} + \frac{1}{3})} dx$$

$$= \frac{1}{3} \int \frac{6x-1}{3x^2-x+1} dx + \frac{10}{3} \cdot \frac{1}{3} \int \frac{1}{x^2 - 2x \cdot \frac{1}{6} + (\frac{1}{6})^2 + \frac{1}{3} - \frac{1}{36}} dx$$

$$= \frac{1}{3} \int \frac{6x-1}{3x^2-x+1} dx + \frac{10}{9} \int \frac{1}{(x - \frac{1}{6})^2 + (\frac{\sqrt{11}}{6})^2} dx$$

$$= \frac{1}{3} \int \frac{6x-1}{3x^2-x+1} dx + \frac{10}{9} \int \frac{dx}{(x - \frac{1}{6})^2 + (\frac{\sqrt{11}}{6})^2}$$

$$= \frac{1}{3} \ln |3x - x + 1| + \frac{10}{9} \cdot \frac{1}{\frac{\sqrt{11}}{6}} \tan^{-1} \frac{x - \frac{1}{6}}{\frac{\sqrt{11}}{6}} + C$$

$$= \frac{1}{3} \ln |3x - x + 1| + \frac{20}{3\sqrt{11}} \tan^{-1} \frac{6x - 1}{\sqrt{11}} + C$$

Ans

Method #05 :-

#Q-6:

let,

$$I = \int \frac{x^{\check{}} - x + 1}{x^{\check{}} + x + 1} dx$$

$$= \int \frac{(x^{\check{}} + x + 1) - 2x}{x^{\check{}} + x + 1} dx$$

$$= \int \frac{(x^{\check{}} + x + 1)}{(x^{\check{}} + x + 1)} dx - \int \frac{2x}{x^{\check{}} + x + 1} dx$$

$$= \int dx - \int \frac{2x}{x^{\check{}} + x + 1} dx$$

$$= x - \int \frac{(2x + 1) - 1}{x^{\check{}} + x + 1} dx$$

$$= x - \int \frac{2x + 1}{x^{\check{}} + x + 1} + \int \frac{1}{x^{\check{}} + x + 1} dx$$

$$= x - \ln |x^{\check{}} + x + 1| + \int \frac{dx}{x^{\check{}} + 2 \cdot x \cdot \frac{1}{2} + (\frac{1}{2})^{\check{}} - \frac{1}{4} + 1}$$

$$= x - \ln(x^{\sqrt{}} + x + 1) + \int \frac{dx}{(x + \frac{1}{2})^{\sqrt{}} + (\frac{\sqrt{3}}{2})^{\sqrt{}}}$$

$$= x - \ln|x^{\sqrt{}} + x + 1| + \frac{1}{\frac{\sqrt{3}}{2}} \tan^{-1} \frac{x + \frac{1}{2}}{\frac{\sqrt{3}}{2}} + C$$

$$= x - \ln|x^{\sqrt{}} + x + 1| + \frac{2}{\sqrt{3}} \tan^{-1} \frac{2x+1}{\sqrt{3}} + C$$

Am

8-7%

let,

$$I = \int \frac{x^{\sqrt{}}}{x^{\sqrt{}} - 4} dx$$

$$= \int \frac{(x^{\sqrt{}} - 4) + 4}{x^{\sqrt{}} - 4} dx$$

$$= \int \frac{(x^{\sqrt{}} - 4)}{(x^{\sqrt{}} - 4)} dx + \int \frac{4}{x^{\sqrt{}} - 4} dx$$

$$= \int dx + 4 \int \frac{1}{x^{\sqrt{}} - 2^{\sqrt{}}} dx$$

$$= x + 4 \cdot \frac{1}{2 \cdot 2} \ln \left| \frac{x-2}{x+2} \right| + C$$

$$= x + \ln \left| \frac{x-2}{x+2} \right| + C$$

Method-7:- $\int \frac{dx}{(ax+b)\sqrt{cx+d}}$ $a \neq 0, c \neq 0, cx+d = z^2$

Q-9:-

Given that,

$$I = \int \frac{dx}{(2+x)\sqrt{1+x}}$$

let,

$$1+x = z^2$$

$$= \int \frac{2z dz}{(2+z^2-1)\sqrt{z^2}}$$

$$\Rightarrow dx = 2z dz$$

Again,

$$1+x = z^2$$

$$= 2 \int \frac{z dz}{(z^2+1)z}$$

$$\Rightarrow x = z^2 - 1$$

$$= 2 \tan^{-1} z + C$$

$$= 2 \tan^{-1} \sqrt{1+x} + C \quad \text{Ans}$$

Q-8:-

Given that,

$$I = \int \frac{dx}{(2x+1)\sqrt{4x+3}}$$

$$= \int \frac{\frac{1}{2} dz}{\left(2 \cdot \frac{z^2-3}{4} + 1\right) \sqrt{z^2}}$$

$$= \frac{1}{2} \int \frac{z dz}{\left(\frac{z^2-3}{2} + 1\right) z}$$

$$= \frac{1}{2} \int \frac{z dz}{\left(\frac{z^2-3+2}{2}\right) z}$$

$$= \frac{1}{2} \int \frac{z dz}{z^2-3+2}$$

$$= \int \frac{dz}{z^2-1^2}$$

$$= \frac{1}{2 \cdot 1} \ln \left| \frac{z-1}{z+1} \right|$$

let

$$4x+3 = z^2$$

$$\Rightarrow 4dx = 2z dz$$

$$\Rightarrow dx = \frac{1}{2} z dz$$

Again,

$$z^2 = 4x+3$$

$$\Rightarrow x = \frac{z^2-3}{4}$$

Ans

Method #13

$$\int u v dx = u \int v dx - \int \left\{ \frac{d}{dx} (u) \int v dx \right\} dx$$

Q-10:-

let,

$$I = \int x^{\vee} e^x dx$$

$$= x^{\vee} \int e^x dx - \int \left\{ \frac{d}{dx} x^{\vee} \int e^x dx \right\}$$

#8-11:-

let,

$$I = \int x \ln x \, dx$$

$$= \ln x \int x \, dx - \int \left\{ \frac{d}{dx} \ln x \int x \, dx \right\} dx$$

$$= \ln x \cdot \frac{x^2}{2} - \int \left\{ \frac{1}{x} dx \int x \, dx \right\} dx$$

$$= \ln x \cdot \frac{x^2}{2} - \int \frac{1}{x} \cdot \frac{x^2}{2} dx$$

$$= \frac{x^2 \ln x}{2} - \frac{1}{2} \int x \, dx$$

$$= \frac{x^2 \ln x}{2} - \frac{x^2}{4} + C \quad \left[\int x^n = \frac{x^{n+1}}{n+1} \right]$$

Ans

8-12:-

let,

$$I = \int \tan^{-1} x \, dx$$

$$= \tan^{-1} x \int dx - \int \left\{ \frac{d}{dx} (\tan^{-1} x) \int dx \right\} dx$$

$$= x \tan^{-1} x - \int \left\{ \frac{1}{1+x^2} \cdot x \right\} dx$$

$$= x \tan^{-1} x - \frac{1}{2} \int \frac{2x}{1+x^2} dx$$

$$= x \tan^{-1} x - \frac{1}{2} \ln |1+x^2| + c$$

Method # 14 :-

$$\int \frac{1}{x} dx = \ln x + c$$

$$\int e^x \{ f(x) + f'(x) \} dx = e^x f(x) + c$$

Q-13:-

Nabil's Note

CSE - 54 Batch

R.S,

$$\int e^x f(x) dx$$

$$= f(x) \int e^x dx - \int \left\{ \frac{d}{dx} f(x) \int e^x dx \right\} dx$$

$$= f(x) e^x - \int f'(x) e^x dx$$

$$= e^x f(x) - \int e^x f'(x) dx$$

$$\therefore \int e^x f(x) dx = e^x f(x) - \int e^x f'(x) dx$$

$$\Rightarrow e^x f(x) = \int e^x f(x) dx + \int e^x f'(x) dx$$

$$\Rightarrow e^x f(x) = \int e^x \{f(x) + f'(x)\} dx + C$$

$$\Rightarrow \int e^x \{f(x) + f'(x)\} dx = e^x f(x) + C$$

[Proved]

#Q-14:

let,

$$I = \int e^x (\sin x - \cos x) dx$$

Here,

$$f(x) = -\cos x$$

$$f'(x) = \sin x$$

Therefore,

$$I = \int e^x \{f'(x) + f(x)\} dx$$

$$= e^x f(x) + C = e^x \cdot -\cos x + C$$

#Q-15:

let, $\int \frac{x e^x}{(x+1)^2} dx$

$$= \int \frac{(x+1) - 1}{(x+1)^2} e^x dx$$

$$= \int \left\{ \frac{x+1}{(x+1)^2} - \frac{1}{(x+1)^2} \right\} e^x dx$$

$$= \int e^x \left\{ \frac{x+1}{(x+1)^2} - \frac{1}{(x+1)^2} \right\} dx$$

$$= \int e^x \left\{ \frac{1}{x+1} + \frac{-1}{(x+1)^2} \right\} dx$$

Hence,

$$f(x) = \frac{1}{x+1}$$

$$f(x) = (x+1)^{-1}$$

$$f'(x) = -\frac{1}{(x+1)^2}$$

$$= -(x+1)^{-2}$$

Therefore,

$$I = \int e^x \{ f(x) + f'(x) \} dx$$

$$= e^x \int (x) + c$$

$$= e^x \cdot \frac{1}{x+1} + c$$

Group B

1. Geometrical Interpretation

Q-1. Evaluate the integral $\int_0^1 x \, dx$ by geometrically.

Q-2. Evaluate the integral $\int_0^1 x^2 \, dx$ by geometrically.

Q-3. Evaluate the integral $\int_0^1 x^3 \, dx$ by geometrically.

2. Reduction Formula

Q-1. Prove that, $\int \sin^n x \, dx = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x \, dx$

Q-2. Prove that, $\int \cos^n x \, dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x \, dx$

Q-3. If $I_n = \int_0^{\pi/4} \tan^n \theta \, d\theta$ and $n > 1$ then prove that, $n(I_{n-1} + I_{n+1}) = 1$

3. Properties of Definite Integral

Q-1. Show that, $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} \, dx = \frac{\pi}{4}$

Q-2. Show that, $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \cos x} \, dx = \frac{\pi}{4}$

Q-3. Show that, $\int_0^{\pi} \cos^{15} x \, dx = 0$

Q-4. Show that, $\int_0^{\pi/2} \frac{(\sin x)^{3/2}}{(\sin x)^{3/2} + (\cos x)^{3/2}} \, dx = \frac{\pi}{4}$

4. Successive Integral

Q-1. Evaluate the multiple integral $\int_{-3}^3 \int_0^1 \int_1^2 (x + y + z) \, dx \, dy \, dz$

Q-2. Evaluate the multiple integral $\int_2^3 \int_0^{3y} \int_1^{yz} (2x + y + z) \, dx \, dz \, dy$

Q-3. Evaluate the triple integral $\int_0^1 \int_0^{1-x} \int_0^{1-y^2} z \, dz \, dy \, dx$

Q-4. Evaluate the triple integral $\int_0^1 \int_0^x \int_0^y x^3 y^2 z \, dz \, dy \, dx$

5. Gamma and Beta Function

Q-1. Define Gamma and Beta function.

Q-2. Prove that, $\Gamma(n+1) = n\Gamma(n)$

Q-3. Prove that, $\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta d\theta$

Q-4. Prove that, $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

Q-5. Show that, $\int_0^{\pi/2} \cos^8 x \sin^6 x dx = \frac{5\pi}{4096}$

Q-6. Prove that, $\int_0^{\pi} x \cos^4 x dx = \frac{3\pi^2}{16}$

6. Integration by Revolution

Arc Length of a Curve

Q-1. Find the circumference of the circle $x^2 + y^2 = a^2$ where a is the radius.

Q-2. Find the arc length along the curve $y = \frac{2}{3}x^{3/2}$ between $x = 3$ and $x = 8$

Areas of Surfaces of Revolution

Q-3. Find the area of the surface of revolution obtained by revolving the graph of $y = 2x$ from $x = -3$ to $x = -1$ about the x-axis.

Q-4. Find the area of the surface generated by revolving the arc of the curve of $y = x^3$ from $x = -1$ to $x = 2$ about the x-axis.

Q-5. Find the area of the surface generated by revolving the arc of the curve of $y = x^2$ from $y = 0$ to $y = 1$ about the y-axis.

Volumes of Solids of Revolution

Q-6. Find the volume of the solid of revolution generated by the graph of $y = x^2$ between $x = 0$ to $x = 2$ about the x-axis.

Q-7. Find the volume generated by the area bounded by the curve $y = 2x - x^2$ and $y = 0$ when they are revolved about the x-axis.

Q-8. Find the volume of the solid of revolution generated by rotating the curve $y = x^3$ between $y = 0$ to $y = 4$ about the y-axis.

1. Geometrical Interpretation

Que 1. Evaluate the integral $\int_0^1 x dx$ by geometrically

Sol: We know that,

$$\int_a^b f(x) dx = \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h \cdot f(a+rh); \text{ where } nh = b-a$$

Now according to question,

$$a = 0$$

also,

$$b = 1$$

$$f(x) = x$$

$$\therefore f(a+rh) = a+rh$$

$$\text{or, } f(rh) = rh$$

Therefore,

$$\int_0^1 x dx = \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h f(0+rh)$$

$$= \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h \cdot rh$$

$$= \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h^2 r$$

$$= \lim_{h \rightarrow 0} h^2 \sum_{r=1}^n r$$

$$= \lim_{h \rightarrow 0} h^2 (1+2+3+\dots+n)$$

$$= \lim_{h \rightarrow 0} h^2 \cdot \frac{n(n+1)}{2}$$

$$= \lim_{h \rightarrow 0} \cdot \frac{1}{2} nh (nh+h)$$

$$= \lim_{h \rightarrow 0} \frac{1}{2} \cdot 1 \cdot (1+h) \left[\begin{array}{l} \text{as } nh = b-a \\ \text{or, } nh = 1-0 = 1 \end{array} \right]$$

$$= \frac{1}{2}$$

(Ans.)

Que 2. Evaluate the integral $\int_0^1 x^2 dx$ by geometrically.

Sol: We know that,

$$\int_a^b f(x) dx = \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h f(a+rh) ; \text{ where, } nh = b-a$$

Now according to the question,

$$a=0 \quad \text{also, } f(x) = x^2$$

$$b=1 \quad \therefore f(a+rh) = (a+rh)^2 = r^2 h^2$$

Therefore,

$$\int_0^1 x^2 dx = \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h \cdot r^2 h^2$$

$$= \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h^3 r^2$$

$$= \lim_{h \rightarrow 0} h^3 \sum_{r=1}^n r^2$$

$$= \lim_{h \rightarrow 0} h^3 (1+4+9+\dots+n^2)$$

$$= \lim_{h \rightarrow 0} h^3 \cdot \frac{1}{6} n(n+1)(2n+1)$$

$$= \frac{1}{6} \lim_{h \rightarrow 0} nh(nh+h) \cdot (2nh+h)$$

$$= \frac{1}{6} \cdot 1 \cdot (1+0) \cdot (2 \cdot 1+0) \left[\begin{array}{l} \text{as } nh = b-a \\ \text{or, } nh = 1-0 = 1 \end{array} \right]$$

$$= \frac{1}{3} \quad (\text{Ans})$$

Que 3. Evaluate the integral $\int_0^1 x^3 dx$ by geometrically.

Sol: We know that,

$$\int_a^b f(x) dx = \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h f(a+rh) ; \text{ where } nh = b-a$$

Now according to the question,

$$a=0 \quad \text{also, } f(x) = x^3$$

$$b=1 \quad \therefore f(a+rh) = (a+rh)^3 = r^3 h^3$$

Therefore,

$$\int_0^1 x^3 dx = \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h \cdot r^3 h^3$$

$$= \lim_{h \rightarrow 0} \sum_{r=0}^{n-1} h^4 r^3$$

$$= \lim_{h \rightarrow 0} h^4 \sum_{r=1}^n r^3$$

$$= \lim_{h \rightarrow 0} h (1+8+27+\dots+n^3)$$

$$= \lim_{h \rightarrow 0} h \left\{ \frac{n(n+1)}{2} \right\}^2$$

$$= \lim_{h \rightarrow 0} \left\{ \frac{nh(nh+h)}{2} \right\}^2$$

$$= \lim_{h \rightarrow 0} \left\{ \frac{1 \cdot (1+h)}{2} \right\}^2 \quad \left[\begin{array}{l} \text{as, } nh = b-a \\ \text{on, } nh = 1-0=1 \end{array} \right]$$

$$= \left(\frac{1+0}{2} \right)^2$$

$$= \frac{1}{4} \quad (\text{Ans})$$

2. Reduction Formula

Que 1. Prove that, $\int \sin^n x = -\frac{1}{n} \sin^{n-1} x \cos x +$

Sol: Let, $\frac{n-1}{n} \int \sin^{n-2} x dx$

$$I = \int \sin^n x dx \quad \text{--- (i)}$$

$$= \int \sin^{n-1} x \cdot \sin x dx$$

$$= \sin^{n-1} \int \sin x - \int \left\{ \frac{d}{dx} (\sin^{n-1} x) \right\} \sin x dx$$

$$= -\sin^{n-1} x \cos x - \int \{ (n-1) \sin^{n-2} x \cos x \cdot (-\cos x) \} dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \cos^2 x dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x (1 - \sin^2 x) dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int (\sin^{n-2} x - \sin^n x) dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x dx - (n-1) \int \sin^n x dx$$

$$\text{on, } I = -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x dx - (n-1) I \quad [\text{from eq (i)}]$$

$$\text{on, } I + nI - I = -\sin^{n-1} x \cos x + \int \sin^{n-2} x dx (n-1)$$

$$\text{on, } I = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x dx$$

[Proved]

Que 2. Prove that, $\int \cos^n x dx = \frac{1}{n} \cos^{n-1} x \sin x +$

$$\frac{n-1}{n} \int \cos^{n-2} x dx$$

Sol: Let,

$$I = \int \cos^n x dx \quad \text{--- (i)}$$

$$= \int \cos^{n-1} x \cdot \cos x dx$$

$$= \cos^{n-1} x \int \cos x dx - \int \left\{ \frac{d}{dx} (\cos^{n-1} x) \right\} \cos x dx$$

$$= \cos^{n-1} x \sin x - \int \{(n-1) \cos^{n-2} x (-\sin x) \cdot \sin x\} dx$$

$$= \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x \sin^2 x dx$$

$$= \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x (1 - \cos^2 x) dx$$

$$= \cos^{n-1} x \sin x + (n-1) \int (\cos^{n-2} x - \cos^n x) dx$$

$$= \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x dx - (n-1) \int \cos^n x dx$$

$$I = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x dx - (n-1) I \quad \left[\begin{array}{l} \text{from} \\ \text{eq (i)} \end{array} \right]$$

$$\text{or, } I + nI - I = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x dx$$

$$\text{or, } I = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x dx$$

[Proved]

Que 3. If $I_n = \int_0^{\pi/4} \tan^n \theta d\theta$ and $n > 1$ then

prove that, $n(I_{n-1} + I_{n+1}) = 1$

Sol: Given that,

$$I_n = \int_0^{\pi/4} \tan^n \theta d\theta \quad \text{--- (i)}$$

$$\therefore I_{n-1} = \int_0^{\pi/4} \tan^{n-1} \theta d\theta \quad \text{--- (ii)}$$

$$\text{also, } I_{n+1} = \int_0^{\pi/4} \tan^{n+1} \theta d\theta$$

$$= \int_0^{\pi/4} \tan^{n-1} \theta \cdot \tan^2 \theta d\theta$$

$$= \int_0^{\pi/4} \tan^{n-1} \theta (\sec^2 \theta - 1) d\theta$$

$$= \int_0^{\pi/4} \tan^{n-1} \theta \sec^2 \theta d\theta - \int_0^{\pi/4} \tan^{n-1} \theta d\theta$$

$$I_{n+1} = \int_0^{\pi/4} \tan^{n-1} \theta \sec^2 \theta d\theta - I_{n-1} \quad [\text{by (ii)}]$$

$$\text{or, } I_{n-1} + I_{n+1} = \int_0^{\pi/4} \tan^{n-1} \theta \sec^2 \theta d\theta$$

$$= \int_0^1 z^{n-1} dz$$

$$= \left[\frac{z^{n-1+1}}{n-1+1} \right]_0^1$$

$$= \frac{1}{n} [1^n - 0^n]$$

$$= \frac{1}{n}$$

$$\therefore n(I_{n-1} + I_{n+1}) = 1 \quad [\text{Proved}]$$

Let,

$$\tan \theta = z$$

$$\therefore \sec^2 \theta d\theta = dz$$

0	0	$\pi/4$
z	0	1

3. Properties of Definite Integral

Que 1. Show that, $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx = \frac{\pi}{4}$

Sol: Let,

$$I = \int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx \quad \text{--- (i)}$$
$$= \int_0^{\pi/2} \frac{\sin(\frac{\pi}{2} - x)}{\sin(\frac{\pi}{2} - x) + \cos(\frac{\pi}{2} - x)} dx$$
$$I = \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x} dx \quad \text{--- (ii)}$$

Now adding (i) & (ii), we get,

$$2I = \int_0^{\pi/2} \frac{\sin x + \cos x}{\sin x + \cos x} dx$$
$$= \int_0^{\pi/2} dx$$
$$= [x]_0^{\pi/2}$$
$$= \frac{\pi}{2}$$

$$\therefore I = \frac{\pi}{4}$$

Hence, $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx = \frac{\pi}{4}$ [showed]

Que 2. Prove that, $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \frac{\pi}{4}$

Sol: Let,

$$I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx \quad \text{--- (i)}$$

$$= \int_0^{\pi/2} \frac{\sqrt{\sin(\pi/2 - x)}}{\sqrt{\sin(\pi/2 - x)} + \sqrt{\cos(\pi/2 - x)}} dx$$

$$I = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx \quad \text{--- (ii)}$$

Now, adding (i) & (ii) we get,

$$2I = \int_0^{\pi/2} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$$
$$= \int_0^{\pi/2} dx$$
$$= [x]_0^{\pi/2}$$
$$= \frac{\pi}{2}$$

$$\therefore I = \frac{\pi}{4}$$

Hence, $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \frac{\pi}{4}$ [Proved]

Que 3. Show that, $\int_0^{\pi} \cos^{15} x dx = 0$

Sol: Let,

$$I = \int_0^{\pi} \cos^{15} x dx \text{ --- (i)}$$

$$= \int_0^{\pi} \cos^{15} (\pi - x) dx$$

$$I = - \int_0^{\pi} \cos^{15} x dx \text{ --- (ii)}$$

Now, adding (i) & (ii) we get,

$$2I = \int_0^{\pi} (\cos^{15} x - \cos^{15} x) dx$$
$$= 0$$

$$\therefore I = 0$$

$$\text{Hence, } \int_0^{\pi} \cos^{15} x dx = 0$$

Que 4. Show that, $\int_0^{\pi/2} \frac{(\sin x)^{3/2}}{(\sin x)^{3/2} + (\cos x)^{3/2}} dx = \frac{\pi}{4}$ [showed]

Sol: Let,

$$I = \int_0^{\pi/2} \frac{(\sin x)^{3/2}}{(\sin x)^{3/2} + (\cos x)^{3/2}} dx \text{ --- (i)}$$

$$= \int_0^{\pi/2} \frac{\{\sin(\pi/2 - x)\}^{3/2}}{\{\sin(\pi/2 - x)\}^{3/2} + \{\cos(\pi/2 - x)\}^{3/2}} dx$$

$$I = \int_0^{\pi/2} \frac{(\cos x)^{3/2}}{(\cos x)^{3/2} + (\sin x)^{3/2}} dx \text{ --- (ii)}$$

Now adding (i) & (ii) we get,

$$2I = \int_0^{\pi/2} \frac{(\sin x)^{3/2} + (\cos x)^{3/2}}{(\sin x)^{3/2} + (\cos x)^{3/2}} dx$$

$$= \int_0^{\pi/2} dx$$

$$= [x]_0^{\pi/2} = \frac{\pi}{2}$$

$$\text{or, } I = \frac{\pi}{4}$$

$$\text{Hence, } \int_0^{\pi/2} \frac{(\sin x)^{3/2}}{(\sin x)^{3/2} + (\cos x)^{3/2}} dx = \frac{\pi}{4}$$

[showed]

4. Successive Integral

Que 1. Evaluate the multiple integral

$$\int_{-3}^3 \int_0^1 \int_1^2 (x+y+z) dx dy dz$$

Sol: Let,

$$I = \int_{-3}^3 \int_0^1 \int_1^2 (x+y+z) dx dy dz$$

$$= \int_{-3}^3 \int_0^1 \left[\frac{x^2}{2} + xy + xz \right]_1^2 dy dz$$

$$= \int_{-3}^3 \int_0^1 \left(2 + 2y + 2z - \frac{1}{2} - y - z \right) dy dz$$

$$= \int_{-3}^3 \left[\frac{3y}{2} + \frac{y^2}{2} + yz \right]_0^1 dz$$

$$= \int_{-3}^3 \left(\frac{3}{2} + \frac{1}{2} + z \right) dz$$

$$= \left[\frac{3z}{2} + \frac{z}{2} + \frac{z^2}{2} \right]_{-3}^3$$

$$= \frac{9}{2} + \frac{3}{2} + \frac{9}{2} + \frac{9}{2} + \frac{3}{2} - \frac{9}{2} = 12 \quad (\text{Ans})$$

Que 2. Evaluate the multiple integral

$$\int_2^3 \int_0^{3y} \int_1^{y^2} (2x+y+z) dx dz dy$$

Sol: Let,

$$I = \int_2^3 \int_0^{3y} \int_1^{y^2} (2x+y+z) dx dz dy$$

$$= \int_2^3 \int_0^{3y} \left[x^2 + xy + xz \right]_1^{y^2} dz dy$$

$$= \int_2^3 \int_0^{3y} (y^2 z^2 + y^2 z + yz^2 - 1 - y - z) dz dy$$

$$= \int_2^3 \left[y^2 \frac{z^3}{3} + y^2 \frac{z^2}{2} + y \frac{z^3}{3} - z - yz - \frac{z^2}{2} \right]_0^{3y} dy$$

$$= \int_2^3 \left[\frac{27y^5}{3} + \frac{9y^4}{2} + \frac{27y^4}{3} - 3y - 3y^2 - \frac{9y^2}{2} \right] dy$$

$$= \left[\frac{27}{3} \times \frac{y^6}{6} + \frac{9}{2} \times \frac{y^5}{5} + \frac{27}{3} \times \frac{y^5}{5} - \frac{3y^2}{2} - \frac{3y^3}{3} - \frac{9}{2} \times \frac{y^3}{3} \right]_2^3$$

$$= \left[\frac{3y^6}{2} + \frac{9y^5}{10} + \frac{9y^5}{5} - \frac{3y^2}{2} - y^3 - \frac{3y^3}{2} \right]_2^3$$

$$= \left(\frac{2187}{2} + \frac{2187}{10} + \frac{2187}{5} - \frac{27}{2} - \frac{81}{2} \right) - \left(96 + \frac{144}{5} + \frac{288}{5} - 6 - 8 - 12 \right)$$

$$= 15 \cdot 12 \cdot 2$$

(Ans)

Que 3. Evaluate the triple integral

$$\int_0^1 \int_0^{1-x} \int_0^{1-y^2} z \, dz \, dy \, dx$$

Sol: Let,

$$I = \int_0^1 \int_0^{1-x} \int_0^{1-y^2} z \, dz \, dy \, dx$$

$$= \int_0^1 \int_0^{1-x} \left[\frac{z^2}{2} \right]_0^{1-y^2} dy \, dx$$

$$= \int_0^1 \int_0^{1-x} \frac{(1-y^2)^2}{2} dy \, dx$$

$$= \frac{1}{2} \int_0^1 \int_0^{1-x} (1 - 2y^2 + y^4) dy \, dx$$

$$= \frac{1}{2} \int_0^1 \left[y - \frac{2y^3}{3} + \frac{y^5}{5} \right]_0^{1-x} dx$$

$$= \frac{1}{2} \int_0^1 \left\{ (1-x) - \frac{2}{3}(1-x)^3 + \frac{1}{5}(1-x)^5 \right\} dx$$

$$= \frac{1}{2} \left[\frac{(1-x)^2}{2(-1)} - \frac{2}{3} \times \frac{(1-x)^4}{4(-1)} + \frac{1}{5} \times \frac{(1-x)^6}{6(-1)} \right]_0^1$$

$$= \frac{1}{2} \left[-\frac{(1-x)^2}{2} + \frac{(1-x)^4}{6} - \frac{(1-x)^6}{30} \right]_0^1$$

$$= \frac{1}{2} \left\{ (-0+0-0) - \left(-\frac{1}{2} + \frac{1}{6} - \frac{1}{30} \right) \right\}$$

$$= \frac{1}{2} \left(\frac{1}{2} - \frac{1}{6} + \frac{1}{30} \right) = \frac{11}{60} \quad (\text{Ans})$$

Que 4. Evaluate the triple integral $\int_0^1 \int_0^x \int_0^y x^3 y^2 z \, dz \, dy \, dx$

Sol: Let,

$$I = \int_0^1 \int_0^x \int_0^y x^3 y^2 z \, dz \, dy \, dx$$

$$= \int_0^1 \int_0^x \left[\frac{x^3 y^2 z^2}{2} \right]_0^y dy \, dx$$

$$= \int_0^1 \int_0^x \frac{x^3 y^4}{2} dy \, dx$$

$$= \frac{1}{2} \int_0^1 \left[\frac{x^3 y^5}{5} \right]_0^x dx$$

$$= \frac{1}{2} \times \frac{1}{5} \int_0^1 x^8 dx$$

$$= \frac{1}{10} \left[\frac{x^9}{9} \right]_0^1$$

$$= \frac{1}{90} \quad (\text{Ans})$$

5. Gamma and Beta Function

Que 1. Define Gamma and Beta function.

Sol: Gamma function: The integral $\int_0^{\infty} x^{n-1} e^{-x} dx$ is called the second eularian integral.

This function is defined for all the positive values of n ($n > 0$) is known as the Gamma function and denoted by Γn .

$$\text{Hence, } \Gamma n = \int_0^{\infty} x^{n-1} e^{-x} dx$$

Beta Function: The integral $\int_0^1 x^{m-1} (1-x)^{n-1} dx$ is called the first eularian integral. This function is defined for all the positive values of m and n ($m > 0, n > 0$) is known as the Beta function and denoted by $\beta(m, n)$.

$$\text{Hence, } \beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx.$$

Que 2. Prove that, $\Gamma(n+1) = n\Gamma(n)$

Sol: We have,

$$\Gamma n = \int_0^{\infty} x^{n-1} e^{-x} dx \quad \text{--- (i)}$$

putting $n = n+1$ in (i) we get,

$$\Gamma(n+1) = \int_0^{\infty} x^{n+1-1} e^{-x} dx$$

$$\text{or, } \Gamma(n+1) = \int_0^{\infty} x^n e^{-x} dx \quad \text{--- (ii)}$$

Now,

$$\int x^n e^{-x} dx = x^n \int e^{-x} dx - \left\{ \frac{d}{dx} (x^n) \int e^{-x} dx \right\} dx$$

$$= -x^n e^{-x} + \int (nx^{n-1} e^{-x}) dx$$

$$= -x^n e^{-x} + n \int x^{n-1} e^{-x} dx$$

$$\text{So, } \int_0^{\infty} x^n e^{-x} dx = \left[-x^n e^{-x} \right]_0^{\infty} + n \int_0^{\infty} x^{n-1} e^{-x} dx$$

$$\text{or, } \Gamma(n+1) = \left[-x^n e^{-x} \right]_0^{\infty} + n\Gamma n \quad \text{--- by (i) \& (ii)}$$

$$= (-\infty^n e^{-\infty} + 0^n e^{-0}) + n\Gamma n$$

$$= n\Gamma n$$

$$\text{Hence, } \Gamma(n+1) = n\Gamma(n)$$

[Proved]

Trizon

Que 3. Prove that, $\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$

Sol: We know,

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

Let,

$$x = \sin^2 \theta$$

$$\therefore dx = 2 \sin \theta \cos \theta d\theta$$

x	1	0
θ	$\pi/2$	0

Therefore,

$$\beta(m, n) = \int_0^{\pi/2} (\sin^2 \theta)^{m-1} (1 - \sin^2 \theta)^{n-1} \cdot 2 \sin \theta \cos \theta d\theta$$

$$= \int_0^{\pi/2} \sin^{2m-2} \theta \cos^{2n-2} \theta \sin \theta \cos \theta d\theta$$

$$\therefore \beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$$

[Proved]

Que 4. Prove that, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

Sol: We know,

$$\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta \quad \text{--- (i)}$$

putting $m=n=\frac{1}{2}$ in (i) we get,

$$\beta(\frac{1}{2}, \frac{1}{2}) = 2 \int_0^{\pi/2} d\theta$$

$$= 2 [\theta]_0^{\pi/2}$$

$$\text{or, } \beta(\frac{1}{2}, \frac{1}{2}) = \pi \quad \text{--- (ii)}$$

again, we know,

$$\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

$$\therefore \beta(\frac{1}{2}, \frac{1}{2}) = \frac{\Gamma(\frac{1}{2}) \Gamma(\frac{1}{2})}{\Gamma(\frac{1}{2} + \frac{1}{2})}$$

$$\text{or, } \beta(\frac{1}{2}, \frac{1}{2}) = \left(\Gamma(\frac{1}{2})\right)^2 \quad \text{--- (iii)}$$

Now from (ii) & (iii) we get,

$$\left(\Gamma(\frac{1}{2})\right)^2 = \pi$$

$$\therefore \Gamma(\frac{1}{2}) = \sqrt{\pi}$$

[Proved]

Que 5. Show that, $\int_0^{\pi/2} \cos^8 x \sin^6 x dx = \frac{5\pi}{4096}$

Sol: Let,

$$I = \int_0^{\pi/2} \cos^8 x \sin^6 x dx$$

We have, $\int_0^{\pi/2} \sin^m x \cos^n x dx = \beta(m, n) = \frac{\Gamma(\frac{m+1}{2}) \Gamma(\frac{n+1}{2})}{2 \Gamma(\frac{m+n+2}{2})}$

So, $I = \beta(6, 8)$

$$= \frac{\Gamma(\frac{6+1}{2}) \Gamma(\frac{8+1}{2})}{2 \Gamma(\frac{6+8+2}{2})}$$

$$= \frac{\Gamma(\frac{7}{2}) \Gamma(\frac{9}{2})}{2 \Gamma 8}$$

$$= \frac{\frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi} \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi}}{2 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}$$

$$= \frac{5\pi}{4096}$$

Hence, $\int_0^{\pi/2} \cos^8 x \sin^6 x dx = \frac{5\pi}{4096}$

[Showed]

Que 6. Prove that, $\int_0^{\pi} x \cos^4 x dx = \frac{3\pi^2}{16}$

Sol: Let,

$$I = \int_0^{\pi} x \cos^4 x dx \quad \text{--- (i)}$$

$$= \int_0^{\pi} (\pi - x) \cos^4 (\pi - x) dx$$

$$= \int_0^{\pi} \pi \cos^4 (\pi - x) dx - \int_0^{\pi} x \cos^4 (\pi - x) dx$$

$$= \pi \int_0^{\pi} \cos^4 x dx - \int_0^{\pi} x \cos^4 x dx$$

$$= \pi \int_0^{\pi} \cos^4 x dx - I \quad \text{--- by (i)}$$

on, $2I = \pi \int_0^{\pi} \cos^4 x dx$

on, $2I = 2\pi \int_0^{\pi/2} \cos^4 x dx$

on, $I = \pi \int_0^{\pi/2} \sin^0 x \cos^4 x dx$

$$= \pi \frac{\Gamma(\frac{0+1}{2}) \Gamma(\frac{4+1}{2})}{2 \Gamma(\frac{0+4+2}{2})}$$

$$= \pi \cdot \frac{\Gamma \frac{1}{2} \Gamma \frac{5}{2}}{2 \Gamma 3}$$

$$= \pi \cdot \frac{\sqrt{\pi} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi}}{2 \cdot 2 \cdot 1}$$

$$= \frac{3\pi^2}{16}$$

$$\text{Hence, } \int_0^{\pi} x \cos^4 x \, dx = \frac{3\pi^2}{16} \quad [\text{Proved}]$$

6. Integration by Revolution

Que 1. Find the circumference of the circle $x^2 + y^2 = a^2$ where a is the radius.

Sol: Given the equation of the circle,

$$x^2 + y^2 = a^2 \quad \text{--- (i)}$$

which center is at $(0,0)$ and radius a .

Now from (i) we get,

$$y = \pm \sqrt{a^2 - x^2}$$

which means that the upper semi-circle

is $y = \sqrt{a^2 - x^2}$ and the lower semi-circle

is $y = -\sqrt{a^2 - x^2}$

Now we will find the upper semi circle's circumference length i.e.

$$y = \sqrt{a^2 - x^2}$$

$$\text{on, } \frac{dy}{dx} = \frac{-2x}{2\sqrt{a^2 - x^2}}$$

$$\text{on, } \left(\frac{dy}{dx}\right)^2 = \frac{x^2}{a^2 - x^2}$$

$$\text{on, } 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{x^2}{a^2 - x^2} = \frac{a^2}{a^2 - x^2}$$

$$\therefore \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \frac{a}{\sqrt{a^2 - x^2}}$$

Hence, the length of the circumference of the upper semi circle about x axis,

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx$$

$$= \int_{-a}^a \frac{a}{\sqrt{a^2 - x^2}} \, dx$$

$$= a \left[\sin^{-1} \frac{x}{a} \right]_{-a}^a$$

Here,

$a = -a \rightarrow$ lower limit

$b = a \rightarrow$ upper limit

$$\begin{aligned}
 &= a \left(\sin^{-1} \frac{a}{a} - \sin^{-1} \frac{-a}{a} \right) \\
 &= a (\sin^{-1} 1 + \sin^{-1} 1) \\
 &= \pi a
 \end{aligned}$$

Therefore, the circumference of the given circle is $2\pi a$. (Ans)

Que 2. Find the arc length along the curve $y = \frac{2}{3}x^{3/2}$ between $x=3$ and $x=8$.

Sol: Given the equation is,

$$y = \frac{2}{3}x^{3/2} \quad \text{--- (i)}$$

$$\begin{aligned}
 \therefore \frac{dy}{dx} &= \frac{2}{3} \cdot \frac{3}{2} \cdot x^{3/2-1} \\
 &= x^{1/2}
 \end{aligned}$$

$$\text{or, } \left(\frac{dy}{dx} \right)^2 = x$$

$$\text{or, } \sqrt{1 + \left(\frac{dy}{dx} \right)^2} = \sqrt{1+x}$$

So, the length of the curve is,

$$\begin{aligned}
 L &= \int_a^b \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx \\
 &= \int_3^8 \sqrt{1+x} dx \quad \left| \begin{array}{l} \text{Here,} \\ a=3 \\ b=8 \end{array} \right. \\
 &= \left[\frac{(1+x)^{3/2}}{3/2} \right]_3^8 \\
 &= \frac{2}{3} (9^{3/2} - 4^{3/2}) \\
 &= \frac{38}{3} \quad \text{(Ans)}
 \end{aligned}$$

Que 3. Find the area of the surface of revolution obtained by revolving the graph of $y=2x$ from $x=-3$ to $x=-1$ about x -axis.

Sol: Given the equation is,

$$y = 2x$$

$$\therefore \frac{dy}{dx} = 2$$

$$\therefore \sqrt{1 + \left(\frac{dy}{dx} \right)^2} = \sqrt{5}$$

Now the required area of graph of $y=2x$ from $x=-3$ to $x=-1$ about x -axis.

$$\begin{aligned} A &= \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\ &= \int_{-3}^{-1} 2\pi \cdot 2x \cdot \sqrt{5} dx \\ &= 4\sqrt{5}\pi \int_{-3}^{-1} x dx \\ &= 4\sqrt{5}\pi \left[\frac{x^2}{2} \right]_{-3}^{-1} \\ &= 2\sqrt{5}\pi (1-9) \\ &= -16\sqrt{5}\pi \end{aligned}$$

(Am.)

Que 4. Find the area of the surface generated by revolving the arc of the curve of $y=x^3$ from $x=-1$ to $x=2$ about the x -axis.

Sol: Given the equation is,
 $y=x^3$

$$\therefore \frac{dy}{dx} = 3x^2$$

$$\therefore \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + 9x^4}$$

Now, the required area of the graph of $y=x^3$ from $x=-1$ to $x=2$ about x -axis.

$$\begin{aligned} A &= \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\ &= \int_{-1}^2 2\pi \cdot x^3 \cdot \sqrt{1 + 9x^4} dx \end{aligned}$$

Here,
 $a=-1$
 $b=2$

$$\begin{aligned} &= \int_{10}^{145} 2\pi \sqrt{z} \cdot \frac{dz}{36} \\ &= \frac{\pi}{18} \left[\frac{z^{1/2+1}}{\frac{1}{2}+1} \right]_{10}^{145} \end{aligned}$$

$$= \frac{\pi}{27} \left[z^{3/2} \right]_{10}^{145}$$

$$= \frac{\pi}{27} \left\{ (145)^{3/2} - (10)^{3/2} \right\}$$

(Am.)

Let, $1+9x^4=z$
 $\therefore 36x^3 dx = dz$
 $\text{or, } x^3 dx = \frac{dz}{36}$

also,

x	2	-1
z	145	10

Que 5. Find the area of the surface generated by revolving the arc of the curve of $y = x^2$ from $y=0$ to $y=1$ about the y -axis.

Sol: Given the equation,

$$y = x^2$$

$$\text{or, } x = \sqrt{y}$$

$$\text{or, } \frac{dx}{dy} = \frac{1}{2\sqrt{y}}$$

$$\text{or, } \sqrt{1 + \left(\frac{dx}{dy}\right)^2} = \sqrt{1 + \frac{1}{4y}}$$

Now, the area of the surface graph of $y = x^2$ between $y=0$ to $y=1$ about the y -axis,

$$A = \int_a^b 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$= \int_0^1 2\pi \sqrt{y} \sqrt{1 + \frac{1}{4y}} dy$$

$$= \pi \int_0^1 2\sqrt{y} \cdot \frac{\sqrt{1+4y}}{2\sqrt{y}} dy$$

$$= \pi \int_0^1 \sqrt{1+4y} dy$$

$$\left| \begin{array}{l} \text{Here,} \\ a=0 \\ b=1 \end{array} \right.$$

$$= \pi \left[\frac{(1+4y)^{3/2}}{3/2} \cdot \frac{1}{4} \right]_0^1$$

$$= \pi \cdot \frac{2}{3} \cdot \frac{1}{4} \left\{ (1+4)^{3/2} - (1+0)^{3/2} \right\}$$

$$= \frac{\pi}{6} [5^{3/2} - 1] \text{ (mm)}$$

Que 6. Find the volume of the solid of revolution generated by the graph of $y = x^2$ between $x=0$ to $x=2$ about the x -axis.

Sol: Given that,

$$y = x^2$$

Now, the required volume of the graph of $y = x^2$ between $x=0$ to $x=2$ about the x -axis,

$$V = \int_a^b \pi y^2 dx$$

$$= \int_0^2 \pi (x^2)^2 dx$$

$$= \pi \int_0^2 x^4 dx$$

$$= \pi \left[\frac{x^5}{5} \right]_0^2$$

$$\left| \begin{array}{l} \text{Here,} \\ a=0 \\ b=2 \end{array} \right.$$

$$= \frac{\pi}{5} \times 2^5$$

$$= \frac{32\pi}{5} \quad (\text{Ans})$$

Que 7. Find the volume generated by the area bounded by the curve $y = 2x - x^2$ and $y = 0$ when they are revolved about the x -axis.

Sol: Given that,

$$y = 2x - x^2 \quad \text{--- (i)}$$

$$y = 0 \quad \text{--- (ii)}$$

from (i) and (ii) we get,

$$2x - x^2 = 0$$

$$\text{or, } x(2 - x) = 0$$

$$\therefore x = 0, 2$$

Now, the required volume of the curve $y = 2x - x^2$ between $x = 0$ to $x = 2$ about x -axis,

$$V = \int_a^b \pi y^2 dx$$

$$= \int_0^2 \pi (2x - x^2)^2 dx$$

Here,

$$a = 0$$

$$b = 2$$

$$= \pi \int_0^2 (4x^2 - 4x^3 + x^4) dx$$

$$= \pi \left[\frac{4x^3}{3} - \frac{4x^4}{4} + \frac{x^5}{5} \right]_0^2$$

$$= \pi \left(\frac{4 \times 2^3}{3} - 2^4 + \right)$$

$$= \frac{16\pi}{15} \quad (\text{Ans})$$

Que 8. Find the volume of the solid of revolution generated by rotating the curve $y = x^3$ between $y = 0$ to $y = 4$ about the y -axis.

Sol: Given that,

$$y = x^3$$

$$\text{or, } x = y^{\frac{1}{3}}$$

$$\therefore x^2 = y^{\frac{2}{3}}$$

Now, the required volume of the curve $y = x^3$ between $y = 0$ to $y = 4$ about the y -axis,

$$V = \int_a^b \pi x^2 dy$$

$$= \int_0^4 \pi y^{2/3} dy$$

$$= \pi \left[\frac{y^{2/3+1}}{2/3+1} \right]_0^4$$

$$= \pi \cdot \frac{3}{5} \left[y^{5/3} \right]_0^4$$

$$= \frac{3\pi}{5} \times (4^{5/3} - 0)$$

$$= \frac{3\pi}{5} \times 4^{5/3} \quad (\text{Ans.})$$

Here,

$$a = 0$$

$$b = 4$$